

GrounUp Enterprise Specification

Phase 15 - Document, Drawing, Specification & Construction Information Management

Version 1.0.0 | Review-ready | Proposed lock: GES-P15-v1.0.0

1. Mission

Create a governed common information environment that ensures construction teams can identify, review, distribute, retrieve and act on the correct information revision while preserving contractual records, evidence and downstream traceability.

2. Architecture Principles

Authoritative current-set control; immutable revision lineage; permission-aware retrieval; checksum/file integrity; metadata governance; mobile/offline field access; evidence-first AI; open APIs; records retention and legal hold; configurable workflows.

3. Cross-Phase Integration

Phase 15 integrates estimating and AI plan reading, projects, CRM/customer records, workforce, finance/change impacts, procurement/submittals, fleet documents, Phase 13 analytics, Phase 14 HSEQ records, and future GIS/survey, machine-control and scheduling phases.

4. Revision & Information Control

Controlled documents use governed numbering, metadata, revision/version state, review/approval, issue purpose, distribution and supersession. Superseded information remains immutable and traceable. Current-set publication is explicit and auditable.

5. RFIs, Submittals & Transmittals

RFIs and submittals link directly to drawings, specifications, correspondence, cost/schedule impacts and decisions. Formal transmittals preserve issue purpose, recipient list, exact revisions and acknowledgement evidence.

6. Search, OCR & AI Plan/Spec Intelligence

The platform supports OCR, full-text and semantic retrieval, metadata proposals, plan/spec extraction, cross-reference detection, conflict detection and grounded Q&A.; AI outputs must identify source documents, revisions, evidence and confidence and remain subject to human review where consequential.

7. Field & Offline Information

Field users receive permission-aware current document/drawing sets with revision banners, offline synchronization, QR/location links and stale-copy detection. Offline changes synchronize through governed conflict handling.

8. Records, Retention & Legal Hold

Retention schedules govern records by category and jurisdiction. Legal holds suspend normal disposition. Deletion/disposition requires governed authorization and preserves evidence.

9. Closeout & Turnover

As-builts, O&M; manuals, warranties, training records and turnover packages are tracked for completeness, review and owner acceptance.

Functional Domains

Document Control

Controlled creation, numbering, metadata, status, revisions, distribution, retention and supersession.

Drawing Management

Drawing registers, sheets, revisions, disciplines, sets, overlays, markups and issued-for-purpose states.

Specification Management

Specification sections, revisions, addenda, requirements, references and drawing/spec coordination.

Submittals

Submittal packages, items, reviewers, due dates, review cycles, dispositions and closeout.

RFIs

RFI creation, routing, response, impact, linked documents, drawings, cost/schedule effects and closure.

Transmittals & Distribution

Formal issue packages, recipients, acknowledgements, purpose-of-issue and distribution evidence.

Correspondence & Records

Letters, notices, meeting records, emails imported as records, decisions and contractual correspondence.

Version & Revision Control

Immutable revision history, comparison, supersession, rollback, effective versions and audit evidence.

Markup & Review

Collaborative markup, comments, redlines, stamps, review sessions, issue resolution and approval.

OCR & Content Extraction

Text extraction, document classification, metadata proposals, table extraction and searchable indexing.

Plan & Spec Intelligence

AI-assisted plan/spec reading, scope extraction, requirement detection, cross-reference and conflict detection.

Document Search & Retrieval

Full-text, metadata, semantic, sheet, section, project, discipline and permission-aware search.

Records Retention & Legal Hold

Retention schedules, disposition, legal holds, immutable evidence and deletion controls.

External Collaboration

Controlled guest access, consultant/subcontractor review, secure sharing and expiring links.

Field Document Access

Mobile/offline current-set access, QR/location links, acknowledgements and sync controls.

BIM/CDE Coordination

Model/document references, common data environment states, issue links and coordination records.

Closeout & Turnover

As-builts, O&M; manuals, warranties, training records, turnover packages and owner acceptance.

Document Intelligence Digital Twin

Current information state across projects, revisions, decisions, relationships and downstream impacts.

Controlled Artifact Summary

Artifact	Count
Requirements	340
Entities	155
Fields	240
Business Rules	65
Validation Rules	65
Workflows	36
Engines	45
Api Resources	55
Ui Surfaces	38
Reports	32
Kpis	45
Tests	340

KPI Registry

P15-KPI-001 - Current Document Compliance - Document Control - Governed metric definition for Current Document Compliance.

P15-KPI-002 - Current Drawing Set Compliance - Drawing Management - Governed metric definition for Current Drawing Set Compliance.

P15-KPI-003 - Metadata Completeness - Specification Management - Governed metric definition for Metadata Completeness.

P15-KPI-004 - Revision Processing Time - Submittals - Governed metric definition for Revision Processing Time.

P15-KPI-005 - Review Cycle Time - RFIs - Governed metric definition for Review Cycle Time.

P15-KPI-006 - Approval Cycle Time - Transmittals & Distribution - Governed metric definition for Approval Cycle Time.

P15-KPI-007 - Submittal On-Time % - Correspondence & Records - Governed metric definition for Submittal On-Time %.

P15-KPI-008 - Submittal Review Cycle Time - Version & Revision Control - Governed metric definition for Submittal Review Cycle Time.

P15-KPI-009 - RFI Response Time - Markup & Review - Governed metric definition for RFI Response Time.

P15-KPI-010 - Open RFI Aging - OCR & Content Extraction - Governed metric definition for Open RFI Aging.

P15-KPI-011 - Transmittal Acknowledgement % - Plan & Spec Intelligence - Governed metric definition for Transmittal Acknowledgement %.

P15-KPI-012 - Distribution Accuracy - Document Search & Retrieval - Governed metric definition for Distribution Accuracy.

P15-KPI-013 - Markup Resolution Time - Records Retention & Legal Hold - Governed metric definition for Markup Resolution Time.

P15-KPI-014 - Superseded Copy Exposure - External Collaboration - Governed metric definition for Superseded Copy Exposure.

P15-KPI-015 - External Share Compliance - Field Document Access - Governed metric definition for External Share Compliance.

P15-KPI-016 - Legal Hold Compliance - BIM/CDE Coordination - Governed metric definition for Legal Hold Compliance.

P15-KPI-017 - Retention Disposition Compliance - Closeout & Turnover - Governed metric definition for Retention Disposition Compliance.

P15-KPI-018 - OCR Success Rate - Document Intelligence Digital Twin - Governed metric definition for OCR Success Rate.

P15-KPI-019 - Classification Accuracy - Document Control - Governed metric definition for Classification Accuracy.

P15-KPI-020 - Metadata Proposal Acceptance - Drawing Management - Governed metric definition for Metadata Proposal Acceptance.

P15-KPI-021 - Search Success Rate - Specification Management - Governed metric definition for Search Success Rate.

P15-KPI-022 - Search Time to Document - Submittals - Governed metric definition for Search Time to Document.

P15-KPI-023 - Document Download Volume - RFIs - Governed metric definition for Document Download Volume.

P15-KPI-024 - Mobile Current-Set Sync - Transmittals & Distribution - Governed metric definition for Mobile Current-Set Sync.

P15-KPI-025 - Offline Sync Success - Correspondence & Records - Governed metric definition for Offline Sync Success.

P15-KPI-026 - Drawing Compare Usage - Version & Revision Control - Governed metric definition for Drawing Compare Usage.

P15-KPI-027 - Specification Compare Usage - Markup & Review - Governed metric definition for Specification Compare Usage.

P15-KPI-028 - Conflict Finding Closure - OCR & Content Extraction - Governed metric definition for Conflict Finding Closure.

P15-KPI-029 - Cross-Reference Finding Closure - Plan & Spec Intelligence - Governed metric definition for Cross-Reference Finding Closure.

P15-KPI-030 - Closeout Completeness - Document Search & Retrieval - Governed metric definition for Closeout Completeness.

P15-KPI-031 - As-Built Completion - Records Retention & Legal Hold - Governed metric definition for As-Built Completion.

P15-KPI-032 - O&M; Manual Completion - External Collaboration - Governed metric definition for O&M; Manual Completion.

P15-KPI-033 - Turnover Acceptance - Field Document Access - Governed metric definition for Turnover Acceptance.

P15-KPI-034 - Document Audit Exceptions - BIM/CDE Coordination - Governed metric definition for Document Audit Exceptions.

P15-KPI-035 - Permission Exceptions - Closeout & Turnover - Governed metric definition for Permission Exceptions.

P15-KPI-036 - Controlled Copy Compliance - Document Intelligence Digital Twin - Governed metric definition for Controlled Copy Compliance.

P15-KPI-037 - Revision Impact Review Completion - Document Control - Governed metric definition for Revision Impact Review Completion.

P15-KPI-038 - Document Alert Closure - Drawing Management - Governed metric definition for Document Alert Closure.

P15-KPI-039 - AI Extraction Acceptance - Specification Management - Governed metric definition for AI Extraction Acceptance.

P15-KPI-040 - AI Answer Evidence Coverage - Submittals - Governed metric definition for AI Answer Evidence Coverage.

P15-KPI-041 - AI Confidence Review Rate - RFIs - Governed metric definition for AI Confidence Review Rate.

P15-KPI-042 - Document Digital Twin Freshness - Transmittals & Distribution - Governed metric definition for Document Digital Twin Freshness.

P15-KPI-043 - Document Storage Growth - Correspondence & Records - Governed metric definition for Document Storage Growth.

P15-KPI-044 - Information Workflow SLA - Version & Revision Control - Governed metric definition for Information Workflow SLA.

P15-KPI-045 - Information Management Adoption - Markup & Review - Governed metric definition for Information Management Adoption.

Engine Registry

P15-ENG-001 - Document Numbering Engine - Deterministic - Provides document numbering engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-002 - Naming Convention Engine - Deterministic - Provides naming convention engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-003 - Revision Control Engine - Deterministic - Provides revision control engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-004 - Supersession Engine - Deterministic - Provides supersession engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-005 - File Integrity Engine - Deterministic - Provides file integrity engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-006 - Metadata Validation Engine - Deterministic - Provides metadata validation engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-007 - Document Classification Engine - Deterministic - Provides document classification engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-008 - OCR Engine - Deterministic - Provides ocr engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-009 - Search Index Engine - Deterministic - Provides search index engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-010 - Semantic Search Engine - Deterministic - Provides semantic search engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-011 - Permission Evaluation Engine - Deterministic - Provides permission evaluation engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-012 - Distribution Engine - Deterministic - Provides distribution engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-013 - Transmittal Engine - Deterministic - Provides transmittal engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-014 - Submittal Routing Engine - Deterministic - Provides submittal routing engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-015 - RFI Routing Engine - Deterministic - Provides rfi routing engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-016 - Review/Approval Engine - Deterministic - Provides review/approval engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-017 - Markup Coordination Engine - Deterministic - Provides markup coordination engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-018 - Drawing Compare Engine - Deterministic - Provides drawing compare engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-019 - Specification Compare Engine - Deterministic - Provides specification compare engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-020 - Cross-Reference Engine - Deterministic - Provides cross-reference engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-021 - Conflict Detection Engine - Deterministic - Provides conflict detection engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-022 - Requirement Extraction Engine - Deterministic - Provides requirement extraction engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-023 - Scope Extraction Engine - Deterministic - Provides scope extraction engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-024 - Table Extraction Engine - Deterministic - Provides table extraction engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-025 - Metadata Proposal Engine - Deterministic - Provides metadata proposal engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-026 - Current-Set Engine - Deterministic - Provides current-set engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-027 - Offline Sync Engine - Deterministic - Provides offline sync engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-028 - Retention Engine - Deterministic - Provides retention engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-029 - Legal Hold Engine - Deterministic - Provides legal hold engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-030 - Disposition Engine - Deterministic - Provides disposition engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-031 - External Sharing Engine - Deterministic - Provides external sharing engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-032 - Watermark/Controlled Copy Engine - Deterministic - Provides watermark/controlled copy engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-033 - BIM/CDE Reference Engine - Deterministic - Provides bim/cde reference engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-034 - Closeout Completeness Engine - Deterministic - Provides closeout completeness engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-035 - Turnover Engine - Deterministic - Provides turnover engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-036 - Revision Impact Engine - Deterministic - Provides revision impact engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-037 - Downstream Dependency Engine - Deterministic - Provides downstream dependency engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-038 - Document KPI Engine - Deterministic - Provides document kpi engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-039 - Document Alert Engine - Deterministic - Provides document alert engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-040 - Evidence Retrieval Engine - Deterministic - Provides evidence retrieval engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-041 - Plan/Spec AI Assistant - AI-assisted - Provides plan/spec ai assistant using governed information, permissions, revision lineage and audit evidence.

P15-ENG-042 - Document Summary Engine - AI-assisted - Provides document summary engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-043 - Document Question Answering Engine - AI-assisted - Provides document question answering engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-044 - AI Confidence Engine - AI-assisted - Provides ai confidence engine using governed information, permissions, revision lineage and audit evidence.

P15-ENG-045 - Document Digital Twin Engine - AI-assisted - Provides document digital twin engine using governed information, permissions, revision lineage and audit evidence.

Workflow Registry

P15-WF-001 - Document Creation & Numbering - Document Control - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-002 - Document Review & Approval - Drawing Management - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-003 - Revision & Supersession - Specification Management - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-004 - Drawing Issue & Distribution - Submittals - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-005 - Drawing Review & Markup - RFIs - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-006 - Specification Revision - Transmittals & Distribution - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-007 - Addendum Processing - Correspondence & Records - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-008 - Submittal Creation & Review - Version & Revision Control - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-009 - Submittal Resubmission - Markup & Review - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-010 - RFI Creation & Routing - OCR & Content Extraction - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-011 - RFI Response & Closure - Plan & Spec Intelligence - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-012 - Transmittal Issue - Document Search & Retrieval - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-013 - Formal Correspondence - Records Retention & Legal Hold - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-014 - Meeting Record & Decision Capture - External Collaboration - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-015 - Bulk Document Import - Field Document Access - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-016 - OCR & Classification - BIM/CDE Coordination - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-017 - Metadata Review - Closeout & Turnover - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-018 - AI Plan/Spec Extraction Review - Document Intelligence Digital Twin - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-019 - Drawing/Spec Conflict Review - Document Control - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-020 - Document Search & Retrieval - Drawing Management - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-021 - External Secure Share - Specification Management - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-022 - Guest Review - Submittals - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-023 - Field Current-Set Publication - RFIs - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-024 - Offline Package Sync - Transmittals & Distribution - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-025 - Legal Hold - Correspondence & Records - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-026 - Retention Disposition - Version & Revision Control - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-027 - BIM/CDE Coordination - Markup & Review - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-028 - As-Built Collection - OCR & Content Extraction - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-029 - O&M; Manual Review - Plan & Spec Intelligence - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-030 - Turnover Package - Document Search & Retrieval - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-031 - Owner Acceptance - Records Retention & Legal Hold - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-032 - Document Change Impact Review - External Collaboration - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-033 - Document Audit - Field Document Access - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-034 - Controlled Copy Distribution - BIM/CDE Coordination - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-035 - Document Digital Twin Refresh - Closeout & Turnover - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

P15-WF-036 - Information Release & Rollback - Document Intelligence Digital Twin - Initiate > Validate > Review > Approve/Respond > Publish/Issue > Acknowledge > Close/Archive

Information Governance

Authoritative revision control, permission-aware retrieval, immutable audit history, retention/legal hold and evidence-grounded AI are mandatory. AI cannot autonomously publish authoritative revisions or issue contractual responses.